

UNIT 02 [Intermediate]

Chapter 01 : Programming.

LESSON 01

Automate Your Truck.

Time Frame: [90 mins]

Lesson Objectives:

By the end of this lesson, students will be able to:

1. Understand the basics of programming using **mBlock**.
2. Write a simple program to **blink an LED**.
3. Learn how to **add a device to mBlock** for controlling hardware.
4. Develop a program to **move a motor**.

Challenge Questions:

1. How can we modify the program to **change the LED blinking speed**?
2. What if we want the **truck to stop after 5 seconds**? How would we change the code?
3. Can we make the motor **change direction using programming**? If so, how?

SEL (Social-Emotional Learning) Relation:

1. **Problem-Solving & Critical Thinking** – Students will learn to break down a task into smaller steps and troubleshoot their code.
2. **Collaboration & Communication** – Working in pairs or groups to brainstorm solutions and share their findings.
3. **Resilience & Growth Mindset** – Encouraging students to debug errors and iterate their programming attempts.

Lesson Plan: Programming with mBlock

1. Engage

Teacher Role:

- Facilitate a discussion on how vehicles can move autonomously.
- Ask students about their experiences with programmable toys or robots.

Expected from Students:

- Share ideas on how to make the truck move on its own.
- Discuss prior knowledge of programming or controlling devices.

Parts of the Book Included:



- Introduction with Laila and Adam discussing making the truck move.

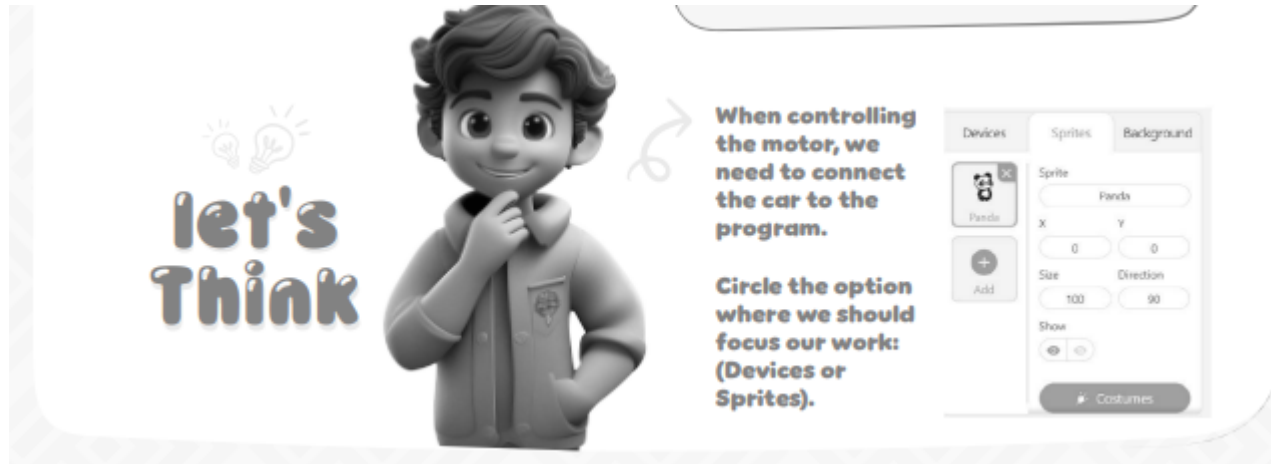
Intro Story:

- Strategy: **Think-Pair-Share** – Have students think about ways to make the truck move, discuss with a partner, then share with the class.

Story Reflection:

- **Questions:**
 1. How do real-world vehicles move without human intervention?
 2. What do we need to make our truck move on its own?
- **Suggested Answers:**
 1. They use programmed instructions and sensors.
 2. We need a program and a control system.
- **Strategy: Visual Brainstorming** – Use a whiteboard to map students' ideas on programming the truck.

2. Think



Questions and Possible Answers:

1. What is programming? (Answer: Giving instructions to a machine.)
2. How can we use programming to make our truck move? (Answer: By writing a program using mBlock.)

Brainstorm Web:

- Students create a web of ideas on what they think is needed for programming.

Strategy: Group Discussion & Concept Mapping – Students create a flowchart of steps to program the truck.

3. Explore

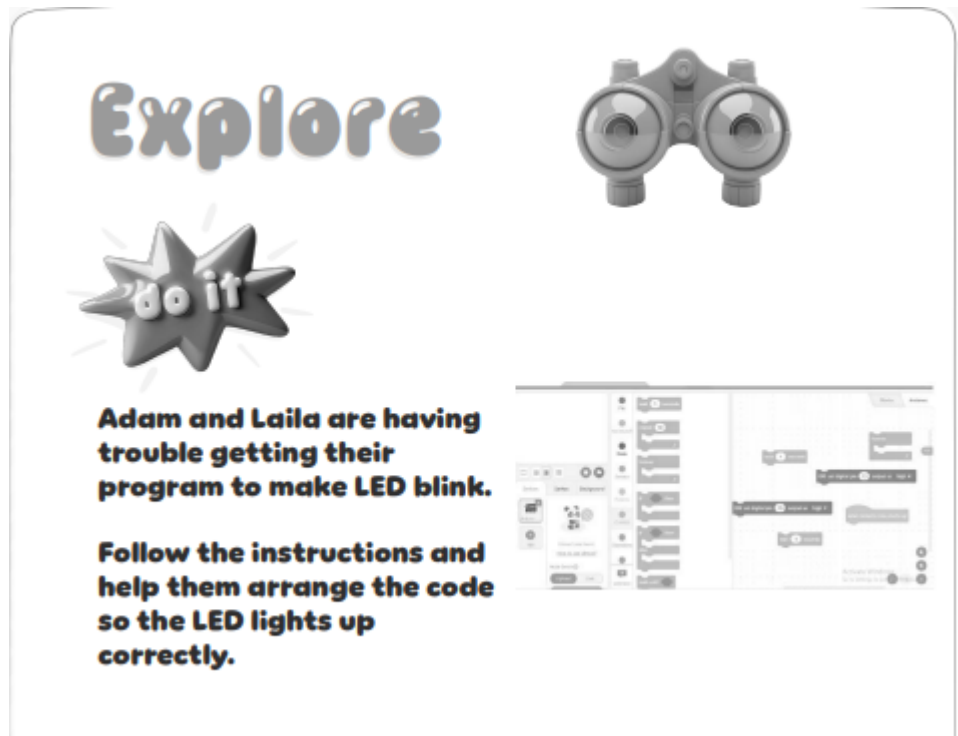
Teacher Role:

- Guide students in setting up mBlock and writing their first program.
- Demonstrate how to make an LED blink as an introduction to programming.

Expected from Students:

- Follow steps to write and execute a basic LED blinking program.
- Test and observe the results.

Parts of the Book Included:



- Activity section with mBlock programming.

Activity - Do It:

- **Materials Needed:**
 - o Computer with mBlock installed
 - o mBlock-compatible microcontroller
 - o LED module
- **Step-by-Step Instructions:**

0. Open mBlock and connect the device.
1. Write a simple program to turn the LED on and off.
2. Upload the program and observe the LED blinking.

- **Reflection Questions:**
 - o What happens when you change the timing in the program?
 - o How do you think we can modify this to make the truck move?

4. Explain

Teacher Role:

- Explain the basic programming concepts of mBlock.
- Guide students in understanding how different commands work.

Expected From Students:

- Engage in discussions and share observations.
- Experiment with modifying the LED program.

Parts of the Book Included:



- Explanation section about programming fundamentals.

Watch It:

- **Pre-Watching:** Discuss what they expect to learn from the video.
- **During Watching:** Encourage note-taking.
- **Post-Watching:** Group discussion on key takeaways.
- **Video Link:** <https://www.youtube.com/watch?app=desktop&v=TefUTnyAIEc>

Read About It:

- **Pre-Reading:** Discuss the importance of programming in automation.
 - **During Reading:** Encourage students to highlight key terms.
 - **Post-Reading:** Summarize and connect to real-world applications.
-

5. Elaborate

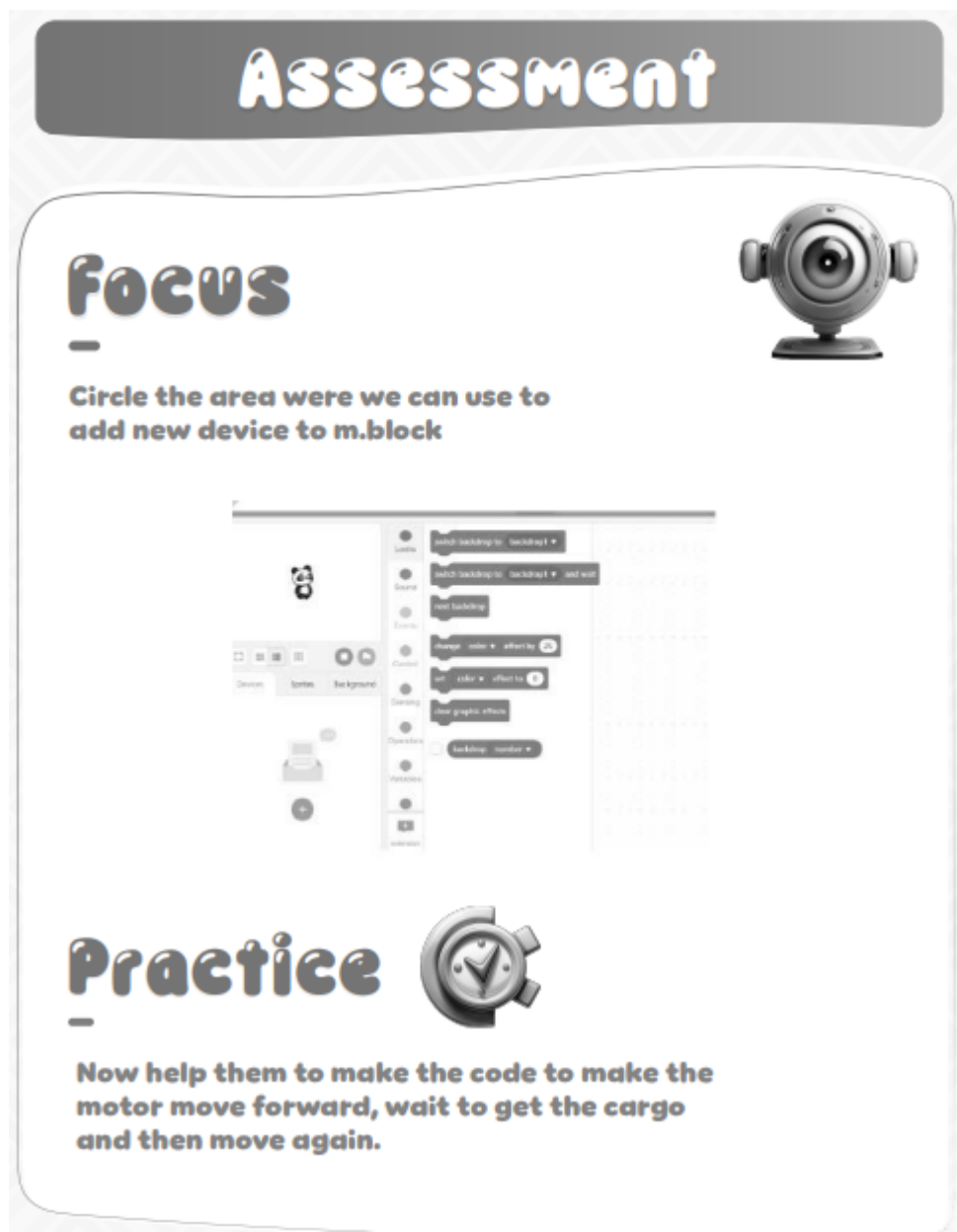
Teacher Role:

- Guide students in adding devices to mBlock.
- Encourage them to think of other applications for programming.

Expected from Students:

- Learn how to add a device to mBlock.
- Try out different programming commands.

Parts of the Book Included:



- Advanced programming techniques section.

Assessment:

- **Focus Questions:**
 1. What command makes the LED blink?
 2. How do we add a device to mBlock?
 3. What is the purpose of programming a moving truck?
- **Practice:**
 - o **Materials Needed:** mBlock software, microcontroller.
 - o **Step-by-Step Instructions:**
 1. Add a new device to mBlock.
 2. Create a simple program for movement.
 - o **Reflection Questions:**
 - How does programming help control real-world machines?

Challenge:

- Try creating a program at home to control another component, such as a buzzer.

6. Evaluate

Teacher Role:

- Assess students' understanding through hands-on projects and discussions.

Expected From Students:

- Demonstrate how they programmed their truck.
- Explain their thought process.

Parts of the Book Included:

- Final project showcase.

Showcase:

- Evaluate students' work using a rubric that assesses:
 - o Program functionality
 - o Creativity in coding solutions
 - o Understanding of programming concepts

Relation to Standards:

- Aligns with STEM and computational thinking standards.
- Encourages problem-solving and logical reasoning skills.

Time Frame: [90 mins]

Lesson Objectives

By the end of this lesson, students will be able to:

1. **Identify and classify** the three types of levers based on the position of the fulcrum, load, and effort.
2. **Explain** how levers work and their role in making tasks easier.
3. **Recognize** the importance of cranes in cargo handling and construction.
4. **Apply** the concept of levers by building a simple crane model.
5. **Engage in hands-on activities** to explore the mechanical advantage of levers.
6. **Reflect on learning** through discussions, assessments, and project presentations.

Challenge Questions

1. **Design Challenge:** If you had to lift a heavy object using only simple machines, how would you combine levers and pulleys to make the task easier? Explain your design.
2. **Real-World Application:** Cranes are used in construction, shipping, and rescue operations. Can you think of another field where cranes or levers play a crucial role? How do they improve efficiency?
3. **Critical Thinking:** Imagine you need to move a large rock using a lever. How would changing the position of the fulcrum affect the effort needed to lift the rock? Why?

Social-Emotional Learning (SEL) through:

1. **Self-Awareness** – Students reflect on their learning by analyzing how levers and cranes function in real life.
2. **Self-Management** – Engages students in hands-on activities, encouraging patience, focus, and problem-solving.
3. **Social Awareness** – Highlights teamwork in engineering and the importance of cranes in transportation and logistics.

Lesson Plan

1- Engage

➤ Teacher Role:

- Facilitate a discussion on simple machines, particularly wheels, axles, and levers.
- Ask guiding questions about cranes and their function in cargo handling.
- Encourage students to share prior knowledge about simple machines.

➤ Expected from Students:

- Participate in discussions by sharing ideas about simple machines.
- Observe the image and make predictions about the lesson's theme.
- Listen to the introductory story and reflect on it.

Stepping Stone:

You need to know...

- ✓ Wheels and axles as simple machine.

You will be able to...

- ✓ Understand Cranes and their role in cargo handling.
- ✓ Understand the principles of levers in different types of machines, like cranes.



Laila

finally we are done with the truck.

Adam

Yes laila, it's time to start working on the crane.

➤ Intro Story:

- **Strategy: "Picture Walk"** – Show the image before reading the story. Ask students what they see and what they think the characters are discussing.
- **Strategy: "Think-Pair-Share"** – After reading the dialogue, ask students to discuss with a partner what they think the story is about.

➤ Story Reflection:

• **Questions:**

1. What are Laila and Adam working on?
2. Why do you think a crane is needed?
3. How do wheels, axles, and levers help in lifting cargo?

• **Suggested Answers:**

1. They are working on a crane.
2. A crane is needed to lift heavy objects efficiently.
3. Wheels and axles help move loads, and levers provide a mechanical advantage for lifting.

- **Strategy: "Role-Playing"** – Have students act out how a crane moves objects using simple props.

- **Strategy: "Interactive Discussion"** – Encourage students to relate the concept to real-life cranes they have seen.
-

2. Think

➤ Questions and Possible Answers

Question: How do cranes make the process of moving and delivering goods faster and easier?

Possible Answers:

- Cranes lift heavy objects that humans cannot lift easily.
- They speed up construction and transportation by moving loads quickly.
- They improve efficiency in ports, warehouses, and construction sites.
- They reduce manual labor and the risk of injuries.

➤ Brainstorm Web

(Provide a blank web diagram with "Cranes" in the center, allowing students to connect ideas such as "Lifting heavy objects," "Faster transportation," "Less manual labor," etc.)

➤ Strategies for Implementation

- **Strategy: "Think-Pair-Share"** – Students think individually about the question, discuss their ideas with a partner, and then share with the class.
 - **Strategy: "Concept Mapping"** – Use a brainstorm web to visually organize how cranes contribute to faster and easier transportation.
 - **Strategy: "Real-World Connection"** – Show images or videos of cranes in action and ask students to explain their importance.
-

3. Explore

Teacher Role:

- Guide students in setting up the experiment with the cups, marbles, and seesaw.
- Explain key concepts such as load, effort, and fulcrum.
- Encourage students to predict what will happen before conducting the activity.
- Ask students to record their observations and reflect on how levers work.

Expected From Students:

- Actively participate in the hands-on experiment.
- Observe and record what happens when marbles are added to the load cup.
- Discuss their observations and relate them to real-world examples.

Parts of the Book Included:

Explore



You have two cups, one called load and the other effort, seesaw and you have group of marbles, put the marbles inside the load cup and write your observation.



Observation:



A lever is a long, strong bar that rests on a support called a fulcrum.

The effort (the force you apply) and the load (the object you're moving) are on either end of the bar.

The fulcrum is the point where the bar turns or pivots.

- "Do It" Activity – Hands-on experiment with cups, marbles, and a seesaw.
- "Watch It" Video – https://www.youtube.com/watch?v=_yzCNniwuw video explaining levers.
- Lever Explanation – Description of load, effort, and fulcrum.

Activity - Do It

Materials Needed:

- Two cups (labeled "Load" and "Effort")
- A small seesaw or balanced stick
- Marbles

Step-by-Step Instructions:

1. Place the seesaw on a stable surface with a pivot point in the middle.
2. Label one cup "Load" and the other "Effort."
3. Place the cups on either end of the seesaw.
4. Start by adding marbles to the "Load" cup and observe what happens.
5. Add force to the "Effort" cup to try and lift the load.
6. Record observations about how the seesaw moves and the role of the fulcrum.

Reflection Questions:

1. What happens when you add marbles to the "Load" cup?
 2. How does the "Effort" cup help lift the load?
 3. What real-life machines use levers to lift heavy objects?
 4. How does changing the position of the fulcrum affect movement?
-

4. Explain

➤ Teacher Role:

- Explain the three classes of levers using visual examples.
- Guide discussions on how different levers work.

➤ Expected from Students:

- Share their observations and relate them to real-world applications.
- Categorize tools into different lever classes.

➤ Parts of the Book Included:

Assessment

focus



Look at each picture and write if it belongs to class 1, class 2, or class 3 based on the position of load, effort, and fulcrum.

CLASS 1	CLASS 2	CLASS 3
7)	8)	9)
1)	2)	3)
4)	5)	6)

➤ Watch It:

- **Pre-Watching:** Ask students to predict what they will learn from the video.
- **During Watching:** Pause and ask guiding questions.
- **Post-Watching:** Students discuss and summarize what they learned.

Strategies for Delivery:

- **Active Listening:** Students take notes while watching.
- **Turn & Talk:** Students share key takeaways with a partner.

➤ Read About It:

- **Pre-Reading:** Discuss prior knowledge of levers.
- **During Reading:** Highlight key points.
- **Post-Reading:** Summarize and answer reflection questions.

Strategies for Delivery:

- **Jigsaw Reading:** Each group reads about a different lever type and presents to the class.
- **Concept Mapping:** Students create a mind map of levers.

5. Elaborate

➤ Teacher Role:

- Encourage students to apply their knowledge creatively.
- Support students in completing the assessment and practice tasks.

➤ Expected from Students:

- Answer assessment questions.
- Engage in hands-on practice.
- Create models of cranes.

➤ Parts of the Book Included:

- **Assessment Section (Lever Classification Activity).**
- **Practice Section (Online Game + Crane Creation).**

➤ Assessment: Focus

- Identify the class of levers in given images.
- Strategies:
 - **Sorting Activity:** Classify real-world tools into lever categories.
 - **Peer Review:** Students compare answers and justify choices.

- o Practice:

Materials Needed:

- Online game (QR code).
- Crane-building materials (craft sticks, pulleys, string).

Step-by-Step Instructions:

1. Play the game to reinforce knowledge of simple machines.
2. Apply the Engineering Design Process (EDP) to build a crane.
3. Test and adjust the crane design.

Reflection Questions:

1. What challenges did you face in building your crane?
2. How did the lever mechanism help lift objects?

➤ **Challenge:**

- Design a lever-based machine at home and present it.

6. Evaluate

➤ **Teacher Role:**

- Assess student understanding through performance tasks.
- Provide constructive feedback.

➤ **Expected from Students:**

- Demonstrate understanding through project presentations.
- Answer reflection questions.

Practice



Let's play the game and find the simple machine.

<https://edheads.org/play-simple-machines/>



Scan Here!

After we learned about levers and we created our crane, let's return to the EDP process and add it to the create part.



Showcase

Create your crane.

Now I can...



Understand Cranes and their role in cargo handling.



Understand the principles of levers in different types of machines, like cranes.

- **Showcase Section:** Students present their crane models.
- Showcase Evaluation:
- **Rubric for Teachers:**
 - o **Creativity (10 points)** – Unique design of the crane.
 - o **Functionality (10 points)** – Proper lever mechanism.
 - o **Presentation (10 points)** – Clear explanation.

➤ Relation to Standards:

- **STEM Integration:** Engineering and design process.
- **Physics Concepts:** Mechanical advantage and force application.

Time Frame: [90 mins]

Lesson Objectives

1. Understand the key components of a crane (fulcrum, effort, and load).
2. Develop creativity by sketching a crane design and translating it into a digital model.
3. Apply engineering principles by adding a pulley system to enhance the crane's functionality.

Challenge Questions

1. How does adding a pulley improve the crane's function?
2. What materials would you use if you were building this crane in real life?
3. How do engineers ensure that cranes remain stable when lifting heavy objects?

SEL Relation

1. **Collaboration & Communication:** Students discuss and share ideas with peers while designing their cranes.
2. **Problem-Solving & Creativity:** Encourages innovative thinking when creating functional designs.
3. **Perseverance & Growth Mindset:** Supports overcoming challenges in digital modeling and refining designs.

Lesson Plan: Designing a Crane

1. Engage

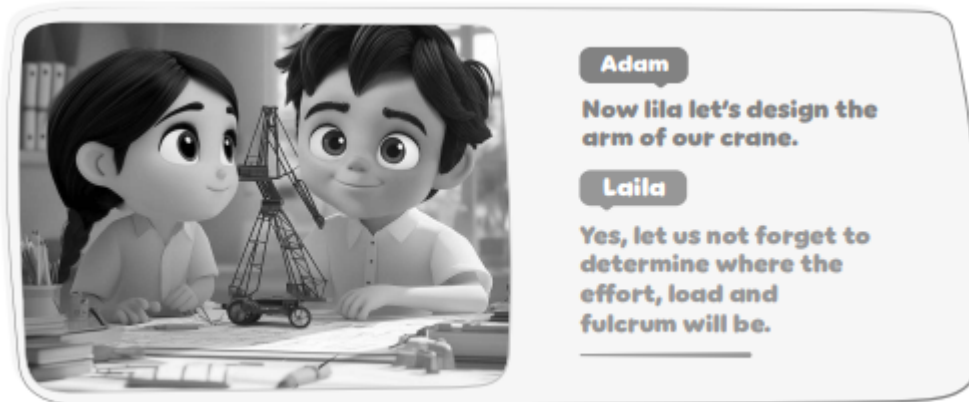
Teacher Role:

- Initiate a discussion about cranes and their uses in construction.
- Ask students what they know about cranes and how they function.
- Introduce the key parts of a crane: fulcrum, effort, and load.

Expected From Students:

- Participate in discussions and share prior knowledge.
- Ask questions and think about how cranes work.

Parts of the Book Included:



- Introduction to simple machines and cranes.

Intro Story:

- Use real-life examples or a short story about construction workers using cranes.
- **Strategy:** Show images/videos of cranes in action and ask students what they observe.

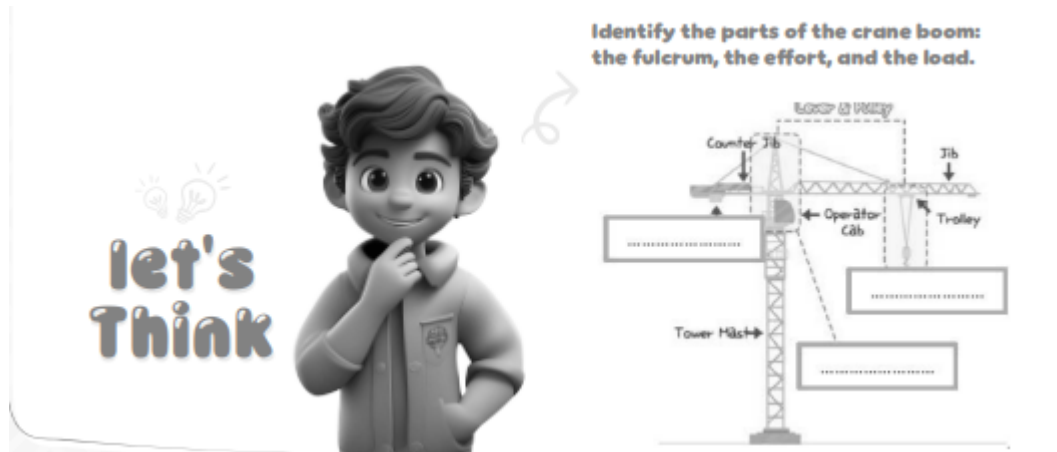
Story Reflection:

- **Questions:**
 - Why do we use cranes?
 - What are the main parts of a crane?
 - How do cranes lift heavy objects?
- **Suggested Answers:**
 - Cranes help move heavy loads in construction.
 - The main parts include the fulcrum, effort, and load.
 - Cranes use pulleys and levers to lift objects.

Strategy: Think-Pair-Share – Students discuss in pairs before sharing with the class.

2. Think

Questions and Possible Answers:



- What simple machines are used in cranes? (*Levers, pulleys, and gears.*)
- How does a crane balance weight? (*By using counterweights and a strong structure.*)

Brainstorm Web:

- Draw a web diagram linking cranes to physics concepts.

Strategy: Use a concept-mapping activity to organize thoughts.

3. Explore

Teacher Role:

- Guide students in visualizing crane designs.
- Encourage creativity in their sketches.

Expected From Students:

- Draw their imagined crane design.
- Label key components like fulcrum, effort, and load.

Parts of the Book Included:

Explore



Picture what your crane boom will look like, and then draw a sketch of it.

Activity - Do It:

- **Materials Needed:** Paper, pencils, rulers, and markers.
- **Step-by-Step Instructions:**
 1. Think about how a crane functions.
 2. Draw a basic structure of a crane.
 3. Label the fulcrum, effort, and load.
- **Reflection Questions:**
 - o How does your crane use simple machines?
 - o What materials would make your crane strong?

4. Explain

Teacher Role:

- Explain how cranes use levers and pulleys to lift loads.
- Show real-life examples through videos.

Expected From Students:

- Listen, observe, and take notes.
- Ask questions about crane mechanics.

Parts of the Book Included:



Watch It:

- **Video:** [How Cranes Work](https://www.youtube.com/watch?v=4581a33lah8)
- **Pre-Watching:** Ask students to predict how cranes function.
- **During Watching:** Take notes on key components.
- **Post-Watching:** Discuss how the video relates to their sketches.

Read About It:

- **Pre-Reading:** Discuss what makes a strong crane.
- **During Reading:** Identify key concepts about crane design.
- **Post-Reading:** Summarize what makes an effective crane design.

5. Elaborate

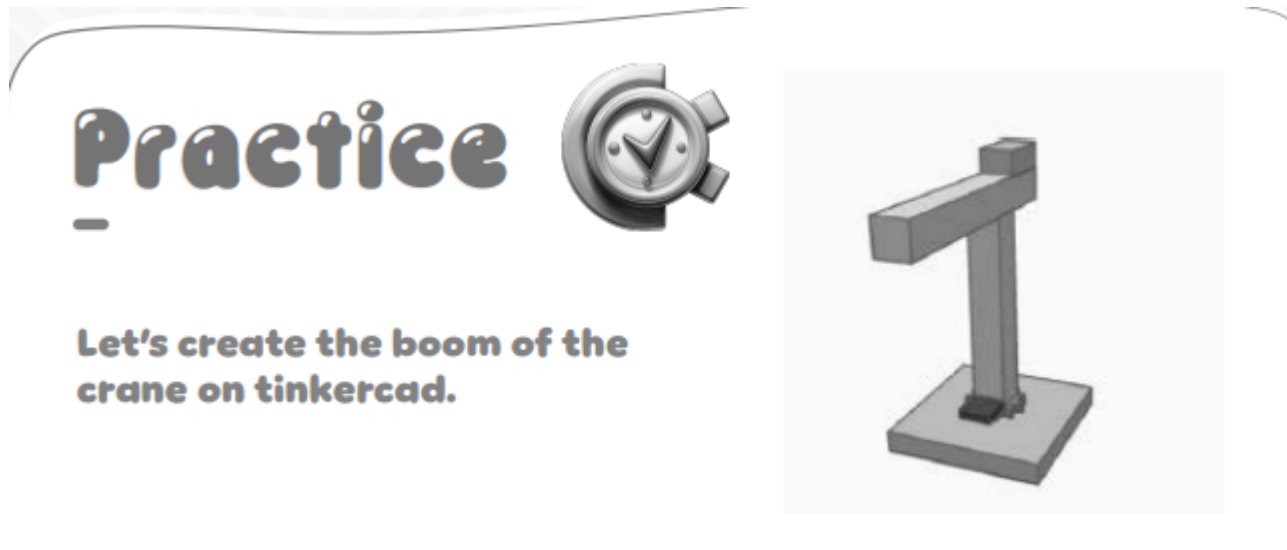
Teacher Role:

- Encourage hands-on learning through digital design.
- Guide students in using Tinkercad.

Expected From Students:

- Apply their knowledge by designing a crane digitally.
- Experiment with different structures.

Parts of the Book Included:



Assessment:

- **Focus:**
 - Draw a crane in Tinkercad.
 - Add labels for fulcrum, effort, and load.
- **Practice:**
 - **Materials Needed:** Computers, internet access, Tinkercad account.
 - **Step-by-Step Instructions:**
 1. Open Tinkercad.
 2. Use basic shapes to create a crane structure.
 3. Label the key components.
 - **Reflection Questions:**
 - How does your digital design compare to your hand-drawn sketch?
 - What changes did you make and why?
- **Challenge:** Add a pulley system to your crane design.

6. Evaluate

Teacher Role:

- Review students' digital crane designs.
- Provide feedback based on a rubric.

Expected From Students:

- Present their crane models.
- Explain their design choices.

Parts of the Book Included:

Showcase

Add the boom to your crane and add pulley to it.

Now I can...

Design the arm of the crane.

Showcase:

- Students display their crane designs and explain the mechanics behind them.